Auditing Audio–Lyrics Variational Autoencoders for Music Clustering

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Abstract

Variational autoencoders (VAEs) can produce compact music representations, but a high internal clustering score does not establish agreement with musical genres. We audit a course project containing spectrograms and lyrics for 683 songs and rebuild its evaluation with ID alignment, train-only preprocessing, grouped held-out splits, and five random seeds. The archived beta-VAE and conditional VAE produce silhouette scores of 0.894 and 0.954 while assigning more than 99% of songs to one cluster; their adjusted Rand indices (ARI) are approximately zero. The rebuilt study compares audio, text, and multimodal representations, regularization strengths, and negative controls using a fixed test set of 102 songs. The convolutional audio–lyrics hybrid reaches ARI 0.207 ± 0.029, compared with 0.096 ± 0.020 for an audio VAE and 0.227 ± 0.028 for text-only features. The results support a benefit over audio alone, but do not establish an advantage over text alone. Two additional grouped splits preserve this ordering. We report uncertainty, reconstruction quality, latent diagnostics and the limitations of the supplied data. A separate English–Bangla experiment addresses the course's bilingual requirement; it is a feasibility study, not evidence of language-invariant genre discovery. The contribution is a reproducible empirical audit and corrected benchmark, rather than a claim of a new VAE architecture.

1 Introduction

Music clustering can support the exploration of collections without training a genre classifier. Audio and lyrics offer different information: acoustic representations describe recorded sound, while lyric representations can reflect vocabulary and subject matter. Their combination is a plausible representation learning strategy, but reconstruction quality, cluster geometry and semantic agreement are distinct objectives. A low-dimensional embedding can be easy to partition without recovering the categories that matter to a listener.

The original course project explored dense, convolutional, multimodal and conditional VAEs. Its archive contains inconsistent data descriptions, preprocessing fitted before splitting, decoder outputs incompatible with some target values, and conclusions that do not follow from the reported metrics. This study repairs those issues and asks three concrete questions: (1) do stronger internal scores correspond to genre agreement; (2) does joint audio–lyrics learning improve on unimodal and linear baselines; and (3) how do KL weighting and modality pairing affect the learned representation?

We provide an executable benchmark with preserved legacy evidence, source hashes, saved splits, fitted transformations and checkpoints. We distinguish fully unsupervised training from a genre-conditioned oracle, and distinguish inductive clustering from within-test descriptive clustering. We also evaluate a small bilingual cohort independently, because the supplied arrays do not establish

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the English–Bangla coverage required by the assignment. These are empirical and reproducibility contributions; we do not claim a novel algorithm or demonstrated disentanglement.

2 Related work

VAEs learn a probabilistic latent representation using an approximate posterior and a reparameterized training objective [6]. A beta-VAE changes the relative strength of prior regularization [4], while free-bits objectives avoid penalizing small per-component KL values [7]. Posterior collapse concerns the relationship between the posterior and prior, not just an imbalanced clustering result [3]. Moreover, unsupervised disentanglement cannot generally be identified without assumptions about models and data [8]. Accordingly, we measure KL, active units and effective rank, and do not infer disentanglement from a visualization.

Multimodal VAEs have been developed with explicit mechanisms for combining modalities and handling missing observations [10]. Our concatenation-based models are simpler and require both inputs; they do not implement product-of-experts inference. Research on lyric topics and genre [1] motivates testing a text-only baseline before crediting joint learning. We use PCA and deterministic autoencoders as audio controls, and TF-IDF with truncated SVD as an auditable text representation. t-SNE [9] is used only for visualization. Two-dimensional separation is not an evaluation criterion.

3 Data and audit

The supplied metadata has 796 rows with track ID, title, audio path, genre and lyrics. Saved processed IDs identify 683 unique rows, each paired with a 128×256 array. The genre counts are rock 178, rap 178, pop 153, country 109 and rb 65. Metadata is joined in the exact processed-ID order. A membership filter happens to preserve this archive's order, but is not an adequate general alignment procedure. Another notebook failed by comparing 683 representations with 796 labels; a separate 500-example synthetic notebook is excluded.

The stored arrays are finite and range from -1 to 1. None of the referenced raw audio files is present. Thus we can reproduce experiments from the supplied features, but cannot independently establish the original sample rate, spectrogram construction, clipping, or normalization. The accompanying source attribution points to a GTZAN Kaggle mirror, and the processed files were received from a collaborator. This does not establish an ID-level mapping to that benchmark. We therefore call this the *supplied corpus*, not a verified GTZAN subset.

The archive also contains a 683×384 lyric embedding matrix without a model revision or row-ID sidecar. Its dimensions do not identify it as BERT. We do not use it in the new primary experiments. Instead, text features are computed from the aligned lyrics. There is no language column and no Bengali-script text in the processed lyrics. Script absence is not a language annotation; multilingual results cannot be inferred from this corpus.

Legacy evidence. Reclustering the saved means reproduces the main pathological scores (Table 1). The beta-VAE K-means partition has sizes $[679, 1, 1, 1, 1]$ and the conditional model has $[678, 1, 1, 1, 2]$. Both have zero active units at the variance threshold used below. Saved posterior variances are unavailable, so this establishes low dispersion and degenerate partitions, not a complete posterior-collapse diagnosis. The old "Spectral" baseline was a second PCA of the same input; it was not an independent spectral baseline.

Table 1: Recomputed legacy K-means results on all 683 saved embeddings, seed 42. These are in-sample diagnostics, not comparable held-out scores.

Saved representation	Silhouette	ARI	Largest share
Conv VAE	0.295	0.044	0.378
Dense VAE	0.202	0.034	0.268
Beta-VAE	0.894	-0.001	0.994
Conditional VAE	0.954	0.000	0.993
PCA	0.103	0.067	0.322

4 Method

4.1 Preprocessing and representations

A fixed 4×4 average pool reduces each supplied spectrogram to 32×64. The pooled values are centered and scaled using a single mean and standard deviation from training data. This makes local CPU experiments tractable and is an explicit change from the original full-resolution models. No physical frequency or time calibration is inferred for the cached arrays.

For lyrics, literal newline escapes are replaced by spaces. A word unigram and bigram TF-IDF vocabulary is learned on training songs only, with sublinear term frequency, minimum document frequency 2, maximum document frequency 0.95, and at most 10,000 terms. Truncated SVD produces 64 features, standardized using the training set. The text-only comparator projects these to 32 dimensions with PCA. The procedure is reproducible and does not require guessing a pretrained transformer identity.

All primary learned embeddings have dimension $d_z = 32$. The dense audio encoder has widths $2048 \to 256 \to 128$, followed by a 128-unit fusion layer and separate mean and log-variance heads. The decoder uses $32 \to 128 \to 256 \to 2048$. Hidden layers use ReLU; output layers are linear because targets are signed continuous features. The convolutional encoder uses three stride-2, 3×3 layers with 8, 16 and 32 channels, producing a $32 \times 4 \times 8$ map, followed by 128 units. Its decoder uses a dense projection and three stride-2 transposed convolutions with 4×4 kernels. The dense multimodal model adds a $64 \to 128 \to 64$ text encoder and concatenates the audio and text branches before the fusion layer. The hybrid replaces the dense audio branch by the convolutional branch. A $32 \to 128 \to 64$ decoder reconstructs text in both multimodal models.

4.2 Objective and regularization

The encoder defines a diagonal Gaussian posterior $q_\phi(z \mid x) = \mathcal{N}(\mu, \mathrm{diag}(\exp \ell))$. Sampling uses $z = \mu + \exp(\ell/2) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. We clamp ℓ to $[-12, 12]$ for numerical stability. Let $D_a = 2048$ and $D_t = 64$. The implemented objective is

$$\mathcal{J} = \mathbb{E}_q\left[\frac{\|x_a - \hat{x}_a\|_2^2}{D_a} + \mathbf{1}_{\mathrm{multimodal}}\frac{\|x_t - \hat{x}_t\|_2^2}{D_t}\right] + \frac{\beta}{d_z}\sum_{j=1}^{d_z}\bar{K}_j, \qquad K_j = \tfrac{1}{2}(\mu_j^2 + e^{\ell_j} - 1 - \ell_j), \tag{1}$$

where $\bar{K}_j$ is the minibatch mean KL for dimension j. This is a dimension-normalized, weighted reconstruction–KL objective, not a reported likelihood bound with a calibrated observation variance. Its beta values are not directly comparable to those of a sum-reconstruction ELBO. Audio-only beta values are $0.001, 0.01, 0.1, 1$. The multimodal, convolutional and hybrid models use 0.01. A free-bits ablation replaces $\bar{K}_j$ by $\max(\bar{K}_j, 0.05)$. The autoencoder decodes μ directly with no sampling or KL penalty.

Controls. PCA retains 32 dimensions of flattened pooled audio. A second baseline uses per-frequency temporal mean and standard deviation, standardized and reduced to 32 dimensions. Direct spectral clustering uses all 2,048 pooled values without PCA. Fusion PCA concatenates audio and text after separately scaling each modality to unit total training variance, then retains 32 components. A negative control independently permutes lyrics within each split before training and evaluation. It preserves text marginals while destroying track pairing. Genre-conditioned audio and multimodal VAEs concatenate a genre one-hot vector at the encoder and audio decoder. They require known genre at test time, so their results are labeled *oracle*, outside the unsupervised comparison.

4.3 Clustering and diagnostics

K-means uses 20 initializations, fits training means, and predicts validation and test assignments. Five clusters are fixed from the course setting and known category count; genre labels otherwise do not enter unsupervised training or model selection. We also report $k \in \{2, 3, 4, 6, 8, 10\}$ without selecting a winner on test labels. Embeddings are centered and divided by one global training RMS norm, which preserves Euclidean relative geometry. We do not whiten each latent dimension and amplify near-constant coordinates.

Ward agglomeration and DBSCAN are fitted directly to test embeddings and are reported as separate *transductive* descriptions, not out-of-sample predictors. DBSCAN uses five neighbors. Radius candidates are the training fifth-neighbor median and 90th percentile times $0.5, 1, 2$, selected by validation coverage times silhouette minus noise fraction; invalid partitions score -2. Noise is excluded from partition metrics and its fraction is always reported. One-cluster, all-noise or singleton-only partitions have undefined internal scores, retained as missing values.

We compute silhouette, Davies–Bouldin, Calinski–Harabasz, ARI, arithmetic NMI, adjusted mutual information, and purity. A majority-class purity baseline, largest-cluster fraction and effective cluster count contextualize the results. Active units satisfy $\mathrm{Var}(\mu_j) > 0.01$. Effective rank is the exponential entropy of the normalized squared singular values of centered means. Internal scores are geometry-specific; their magnitudes across different representations are not calibrated measures of semantic usefulness.

5 Experiments

Duplicate groups join equal normalized titles, normalized lyrics, or exact spectrogram arrays. This yields 676 groups and a fixed group-disjoint split (seed 2026) of 478 training, 103 validation and 102 test songs. Genre is not used to stratify the split. Artist identifiers are unavailable; artist or recording-family independence cannot be claimed. All transformations, including vocabulary, SVD, scalers and PCA, fit only training rows.

We use CPU PyTorch on an Apple M1 MacBook Air (8 GB RAM), four computation threads, Adam with learning rate 10^{-3}, batch size 64, gradient norm clipping at 5, and at most 100 epochs. KL weight increases linearly over 20 epochs. Checkpoint selection starts at epoch 20 and minimizes the full-weight validation objective, with patience 15. Validation uses four fixed Gaussian samples per song, shared across epochs for a model. Training seeds are 11, 22, 33, 44 and 55. We report sample standard deviations across runs on the same split; these do not measure between-dataset uncertainty. The 60 trained neural-model runs in the primary study and requirement extension took 6.4 minutes of summed local training wall time, excluding environment setup, clustering, reporting and the separate bilingual study. Selected epochs ranged from 25 to 100; parameter counts ranged from 206,041 to 1,184,320. The additional split-sensitivity study trained 60 neural models in 7.9 minutes of summed training wall time; preprocessing and evaluation are additional.

Bilingual course experiment. We obtain 20 explicitly English-labeled recordings from JamendoLyrics [2] and 20 Bengali clips from BanglaBeats [5]. Repository revisions and file hashes are pinned. English excerpts start at the first annotated sung line and last three seconds; Bengali excerpts are the released three-second clips, selected at spaced numeric file ranks across eight genres. A shared new recipe uses mono 22,050-Hz audio, FFT size 2,048, hop 512, 128 mel bins, per-clip relative dB clipped at -80 dB, and linear time interpolation to 256 frames, followed by the same pooling. A fixed language-stratified split uses 32 clips for fitting and eight for validation, with an eight-dimensional VAE and PCA. This separate course demonstration fits and evaluates K-means on all 40 representations; it is explicitly transductive and makes no held-out performance claim. BanglaBeats does not provide original song IDs in this release, so distinct-song independence cannot be verified. It also provides no matched lyrics; the multimodal study remains on the supplied corpus.

6 Results

6.1 Held-out genre agreement

Table 2 shows a hybrid ARI of 0.207 ± 0.029 against 0.096 ± 0.020 for the audio VAE at $\beta = 0.01$ and 0.086 ± 0.000 for audio PCA. The paired hybrid-minus-audio-VAE difference is 0.111, with an exploratory 95% bootstrap interval [0.030, 0.195]. Text-only LSA obtains 0.227 ± 0.028 ARI. The hybrid-minus-text difference is -0.020, interval [-0.126, 0.092], so these data do not establish a hybrid advantage over text. Test-set majority-class purity is 0.304. The comparison concerns a fixed corpus and split, not general superiority of one architecture.

Table 2: Inductive K-means on held-out songs, $k = 5$. Mean $\pm$ sample SD over five seeds; higher is better for the displayed metrics.

Representation	ARI	NMI	Silhouette	Purity
Direct spectral	0.086 ± 0.000	0.128 ± 0.000	0.157 ± 0.000	0.441 ± 0.000
Audio PCA	0.086 ± 0.000	0.128 ± 0.000	0.236 ± 0.000	0.441 ± 0.000
Audio statistics	0.084 ± 0.003	0.121 ± 0.008	0.264 ± 0.003	0.449 ± 0.004
Lyrics LSA	0.227 ± 0.028	0.258 ± 0.017	0.056 ± 0.012	0.541 ± 0.013
Fusion PCA	0.089 ± 0.010	0.119 ± 0.005	0.236 ± 0.006	0.447 ± 0.005
Audio AE	0.072 ± 0.010	0.119 ± 0.023	0.292 ± 0.024	0.422 ± 0.030
Audio VAE (0.01)	0.096 ± 0.020	0.128 ± 0.017	0.218 ± 0.016	0.449 ± 0.021
Conv VAE	0.068 ± 0.007	0.103 ± 0.016	0.226 ± 0.030	0.418 ± 0.021
Multimodal VAE	0.153 ± 0.096	0.177 ± 0.094	0.195 ± 0.038	0.467 ± 0.064
Hybrid VAE	0.207 ± 0.029	0.241 ± 0.037	0.149 ± 0.041	0.516 ± 0.019

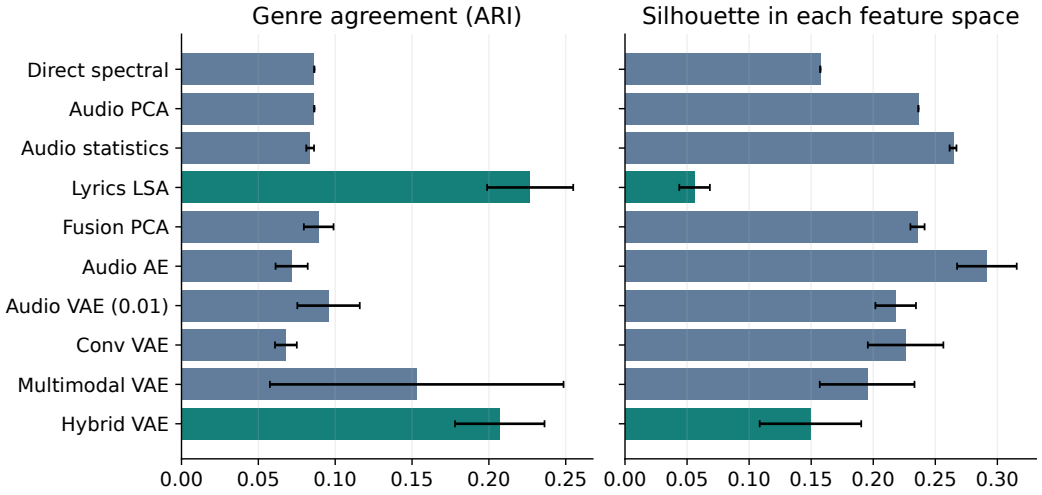


Figure 1: Held-out genre agreement (ARI) and within-representation silhouette. Bars show means over five seeds; error bars show sample standard deviation. Semantic and geometric rankings need not agree. Oracle models are excluded.

Split sensitivity. Two follow-up group-disjoint splits (seeds 2027 and 2028) repeat six neural models and four baselines with the same five training seeds and unchanged hyperparameters (Table 10). The hybrid-minus-audio-VAE mean ARI differences across splits 2026, 2027 and 2028 are +0.111, +0.090, +0.140; hybrid-minus-text differences are -0.020, -0.085, -0.089. These follow-up evaluations assess split sensitivity after inspecting the primary results; they are not independent dataset replications or confirmatory tests.

6.2 Regularization and modality controls

Table 3: Regularization and pairing controls on the same held-out split. FB denotes a 0.05-nat free-bits threshold per latent dimension.

Representation	ARI	Silhouette	Largest share
Audio VAE (0.001)	0.070 ± 0.010	0.247 ± 0.009	0.376 ± 0.026
Audio VAE (0.01)	0.096 ± 0.020	0.218 ± 0.016	0.329 ± 0.016
Audio VAE (0.1)	0.067 ± 0.018	0.251 ± 0.016	0.373 ± 0.048
Audio VAE (1)	0.067 ± 0.012	0.435 ± 0.023	0.363 ± 0.018
Multimodal VAE	0.153 ± 0.096	0.195 ± 0.038	0.406 ± 0.064
Multimodal + FB	0.172 ± 0.078	0.197 ± 0.055	0.402 ± 0.044
Shuffled text	0.032 ± 0.012	0.149 ± 0.024	0.480 ± 0.105

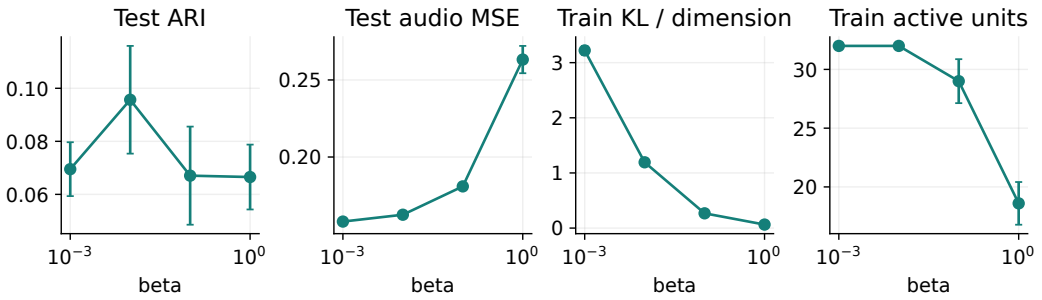


Figure 2: Audio VAE regularization sweep: test genre agreement, test reconstruction MSE at the posterior mean, training KL per dimension and training active units. All panels use five-seed means with sample standard deviations.

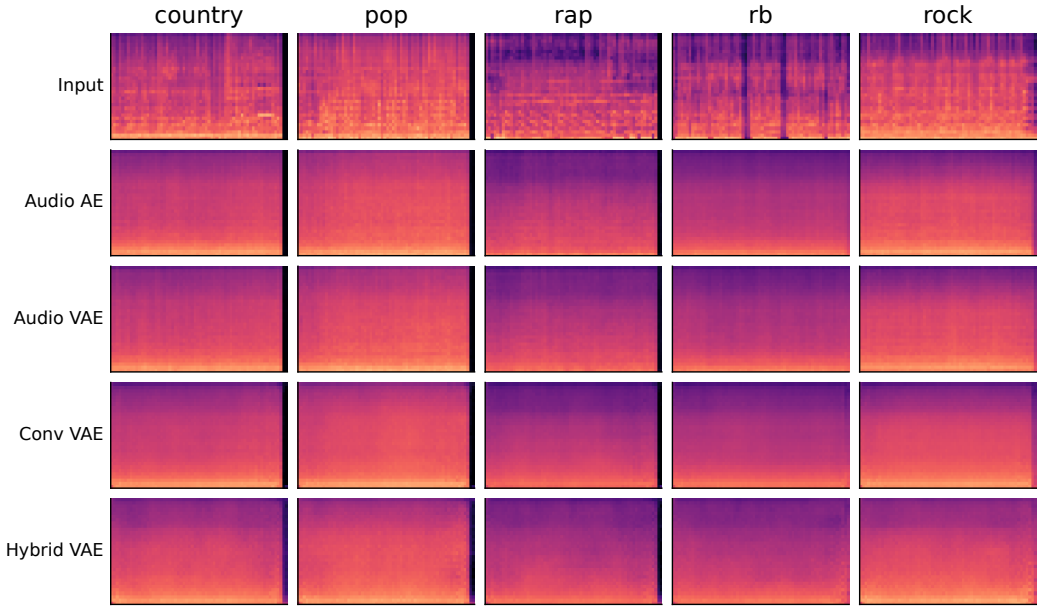


Figure 3: Pooled inputs and posterior-mean reconstructions for the first test example in each genre, using seed 11. Every panel uses the same normalized value range of $[-3, 3]$. Reconstructed coarse structure does not demonstrate perceptual audio quality or genre disentanglement.

The dense multimodal model obtains ARI 0.153 ± 0.096, whereas shuffled pairing obtains 0.032 ± 0.012. The negative control indicates that preserving the relation between a song and its lyrics matters in this setup. Free bits gives 0.172 ± 0.078, but the large run variation prevents a robust superiority claim. Raising audio β to 1 increases mean silhouette to 0.435 while ARI remains 0.067. Figure 2 reports the associated latent and reconstruction changes; a higher geometry score alone is not evidence of improvement.

6.3 Reconstructions and latent visualization

The reconstructed arrays show how models allocate capacity to the cached features (Figure 3). The raw waveforms are unavailable; we do not claim waveform reconstruction or perceptual quality. Appendix plots show test t-SNE projections, genre contingency tables, cluster sizes, interpolation and k sensitivity. A t-SNE fit is descriptive and is never used to compute the reported clustering metrics.

6.4 Bilingual feasibility results

Table 4: Small bilingual demonstration, $k = 5$, mean $\pm$ SD. K-means fits and scores all 40 clips, including training clips. ARI compares language, which equals source.

Representation	Silhouette	CH	Language ARI
PCA	0.175 ± 0.026	10.025 ± 0.222	0.003 ± 0.010
VAE	0.376 ± 0.023	54.907 ± 5.140	-0.005 ± 0.011

Both languages pass through a common audio preprocessing and clustering implementation. Table 4 reports internal scores and language agreement for all clips. Language is perfectly confounded with dataset source and may also covary with genre, recording practice and excerpt selection. These numbers therefore demonstrate execution on bilingual inputs; they do not establish language recognition, language-invariant music structure or generalization to new songs. The appendix plots actual language counts in each cluster.

7 Discussion and limitations

The original "VAE always outperforms PCA" conclusion is not supported. High silhouette can coexist with near-chance genre agreement, and reconstructing both modalities does not automatically make their joint representation more useful than text alone. Genre is also only one imperfect description of music; low ARI does not exclude grouping by another meaningful attribute, but evidence for such attributes requires annotations or a listening study.

The main corpus is small and provenance is incomplete. The primary test set contains only six rb examples. Two additional grouped splits assess sensitivity, but all three reuse the same corpus. Five optimization seeds and conditional bootstrap intervals cannot substitute for independent datasets or artist-disjoint evaluation. The vocabulary/SVD representation is an interpretable baseline rather than a modern pretrained music or multilingual language model. Fixed pooling discards resolution. Cross-model reconstruction objectives differ when text is added, so total losses are not used to rank unimodal against multimodal models. The bilingual cohort is small and its language labels are confounded with collection source. It addresses an implementation requirement, not a causal claim about language.

Data licenses and original extraction provenance for the supplied corpus remain unresolved. The local package contains research artifacts, but is not evidence of permission to redistribute songs or lyrics. Public release should provide source-specific access instructions and verified licensing. Model weights and reconstructions can retain information about the training material. Broader applications would also require evaluation of language and genre coverage to avoid presenting an uneven collection as culturally representative.

A conference submission requires a clearer generalizable finding, replication on documented datasets, suitable pretrained audio/text baselines, and review of any target venue's current format and policies. The present repaired benchmark is a basis for that work, not a claim that a top-tier submission is ready.

8 Conclusion

This study repairs a music VAE project by connecting every reported result to an executable, ID-aligned evaluation. It replaces unsupported multilingual and transformer claims with explicit data provenance, separates genre-conditioned oracles from unsupervised models, and demonstrates why cluster geometry must be interpreted alongside semantic agreement and balance. The measured comparisons establish a reproducible baseline for subsequent research; stronger generalization and language-invariance claims require further data.

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AI assistance and author review

An AI coding assistant was used to audit the supplied files, implement and run experiments, generate figures, check references, and revise this manuscript. The named author must independently review the code, results, dataset rights and scientific claims before submission. No human listening study, new annotation study, or independent author verification is claimed here.

A Reproduction and full results

The repository README provides environment setup and exact commands. `configs/main.json` specifies the frozen primary study. `results/rebuilt_v1/manifest.json` records package versions, input hashes, source hashes and split counts. Training histories, selected checkpoints, embeddings, predictions and fitted transformations are saved by method and seed. The requirement extension adds direct spectral clustering and a multimodal genre oracle after the actual assignment brief was supplied; it uses the identical split. No test scores were used to change primary training hyperparameters. Bootstrap analyses are exploratory and conditional on this split, not corrected confirmatory hypothesis tests.

Table 5: Additional primary metrics, mean ± SD; DB is lower-better, CH and purity are higher-better. Label-conditioned oracles are shown separately.

Representation	DB	CH	Purity
Audio PCA	1.366 ± 0.000	30.657 ± 0.000	0.441 ± 0.000
Lyrics LSA	3.255 ± 0.357	8.775 ± 0.753	0.541 ± 0.013
Fusion PCA	1.327 ± 0.027	33.499 ± 0.196	0.447 ± 0.005
Audio statistics	1.418 ± 0.026	38.519 ± 0.186	0.449 ± 0.004
Direct spectral	1.809 ± 0.000	18.320 ± 0.000	0.441 ± 0.000
Audio AE	1.431 ± 0.254	29.766 ± 2.691	0.422 ± 0.030
Audio VAE (0.001)	1.573 ± 0.089	24.867 ± 2.106	0.406 ± 0.005
Audio VAE (0.01)	1.607 ± 0.140	19.904 ± 2.621	0.449 ± 0.021
Audio VAE (0.1)	1.503 ± 0.130	22.277 ± 2.453	0.420 ± 0.024
Audio VAE (1)	0.843 ± 0.058	132.305 ± 14.007	0.427 ± 0.015
Conv VAE	1.582 ± 0.093	22.702 ± 3.373	0.418 ± 0.021
Multimodal VAE	1.589 ± 0.176	24.059 ± 3.916	0.467 ± 0.064
Hybrid VAE	1.657 ± 0.129	17.110 ± 3.880	0.516 ± 0.019
Multimodal + FB	1.606 ± 0.279	23.630 ± 4.698	0.476 ± 0.059
Shuffled text	1.788 ± 0.080	17.903 ± 2.682	0.373 ± 0.012

Table 6: Training latent diagnostics and held-out reconstruction MSE at the posterior mean. AU counts variance above 0.01.

Representation	AU	KL/dim	Test MSE
Audio VAE (0.001)	32.000 ± 0.000	3.222 ± 0.082	0.158 ± 0.002
Audio VAE (0.01)	32.000 ± 0.000	1.193 ± 0.068	0.163 ± 0.001
Audio VAE (0.1)	29.000 ± 1.871	0.268 ± 0.017	0.181 ± 0.002
Audio VAE (1)	18.600 ± 1.817	0.064 ± 0.003	0.263 ± 0.009
Conv VAE	32.000 ± 0.000	1.408 ± 0.046	0.157 ± 0.002
Multimodal VAE	32.000 ± 0.000	2.720 ± 0.086	0.183 ± 0.003
Hybrid VAE	32.000 ± 0.000	3.379 ± 0.178	0.194 ± 0.013
Multimodal + FB	32.000 ± 0.000	2.674 ± 0.104	0.183 ± 0.001
Shuffled text	32.000 ± 0.000	2.833 ± 0.168	0.183 ± 0.004

Table 7: Within-test clustering (not an inductive comparison). Coverage is the fraction not labeled noise. Undefined scores are shown as –; when only some runs are valid, metrics average valid runs.

Representation / method	Coverage	Clusters	ARI	Silhouette
Audio PCA Ward	1.000 ± 0.000	5.000 ± 0.000	0.045 ± 0.000	0.233 ± 0.000
Audio PCA DBSCAN	0.490 ± 0.000	2.000 ± 0.000	0.086 ± 0.000	0.412 ± 0.000
Lyrics LSA Ward	1.000 ± 0.000	5.000 ± 0.000	0.176 ± 0.000	0.092 ± 0.000
Lyrics LSA DBSCAN	0.931 ± 0.000	1.000 ± 0.000	0.000 ± 0.000	–
Conv VAE Ward	1.000 ± 0.000	5.000 ± 0.000	0.072 ± 0.007	0.248 ± 0.017
Conv VAE DBSCAN	0.712 ± 0.234	2.000 ± 1.414	0.010 ± 0.015	0.365 ± 0.010
Hybrid VAE Ward	1.000 ± 0.000	5.000 ± 0.000	0.171 ± 0.055	0.159 ± 0.020
Hybrid VAE DBSCAN	0.514 ± 0.354	2.000 ± 1.000	0.121 ± 0.176	0.284 ± 0.071
Multimodal VAE Ward	1.000 ± 0.000	5.000 ± 0.000	0.144 ± 0.028	0.194 ± 0.010
Multimodal VAE DBSCAN	0.698 ± 0.097	2.600 ± 0.548	0.059 ± 0.038	0.270 ± 0.045

Table 8: Exploratory paired ARI differences: 1,000 resamples of test duplicate groups and seeds; percentile intervals conditional on this one split. No multiple-comparison adjustment or model refitting.

Comparison (A minus B)	Difference	95% interval
Hybrid VAE - Audio VAE (0.01)	0.111	[0.030, 0.195]
Hybrid VAE - Lyrics LSA	-0.020	[-0.126, 0.092]
Hybrid VAE - Fusion PCA	0.118	[0.027, 0.211]
Multimodal VAE - Shuffled text	0.122	[0.028, 0.243]
Multimodal + FB - Multimodal VAE	0.019	[-0.060, 0.095]
Audio VAE (0.01) - Audio PCA	0.009	[-0.040, 0.060]

Table 9: Genre-conditioned diagnostic models. Both receive true genre at inference and cannot be interpreted as unsupervised methods.

Oracle	ARI	NMI	Silhouette
Audio oracle	0.067 ± 0.014	0.104 ± 0.015	0.180 ± 0.025
Multimodal oracle	0.164 ± 0.080	0.215 ± 0.079	0.201 ± 0.034

Table 10: Split sensitivity: held-out K-means ARI at $k = 5$, mean $\pm$ sample SD over five training seeds per split. Split 2026 is primary; 2027 and 2028 are follow-up evaluations with unchanged settings. These are overlapping splits of the same corpus.

Representation	Split 2026	Split 2027	Split 2028
Audio PCA	0.086 ± 0.000	0.064 ± 0.006	0.001 ± 0.002
Audio statistics	0.084 ± 0.003	0.102 ± 0.016	0.017 ± 0.001
Lyrics LSA	0.227 ± 0.028	0.277 ± 0.033	0.272 ± 0.031
Fusion PCA	0.089 ± 0.010	0.072 ± 0.004	0.014 ± 0.001
Audio AE	0.072 ± 0.010	0.059 ± 0.041	0.024 ± 0.016
Audio VAE (0.01)	0.096 ± 0.020	0.102 ± 0.012	0.044 ± 0.013
Conv VAE	0.068 ± 0.007	0.084 ± 0.017	0.012 ± 0.005
Multimodal VAE	0.153 ± 0.096	0.178 ± 0.057	0.115 ± 0.044
Hybrid VAE	0.207 ± 0.029	0.192 ± 0.028	0.183 ± 0.042
Shuffled text	0.032 ± 0.012	0.025 ± 0.018	-0.009 ± 0.003

B Additional visualizations

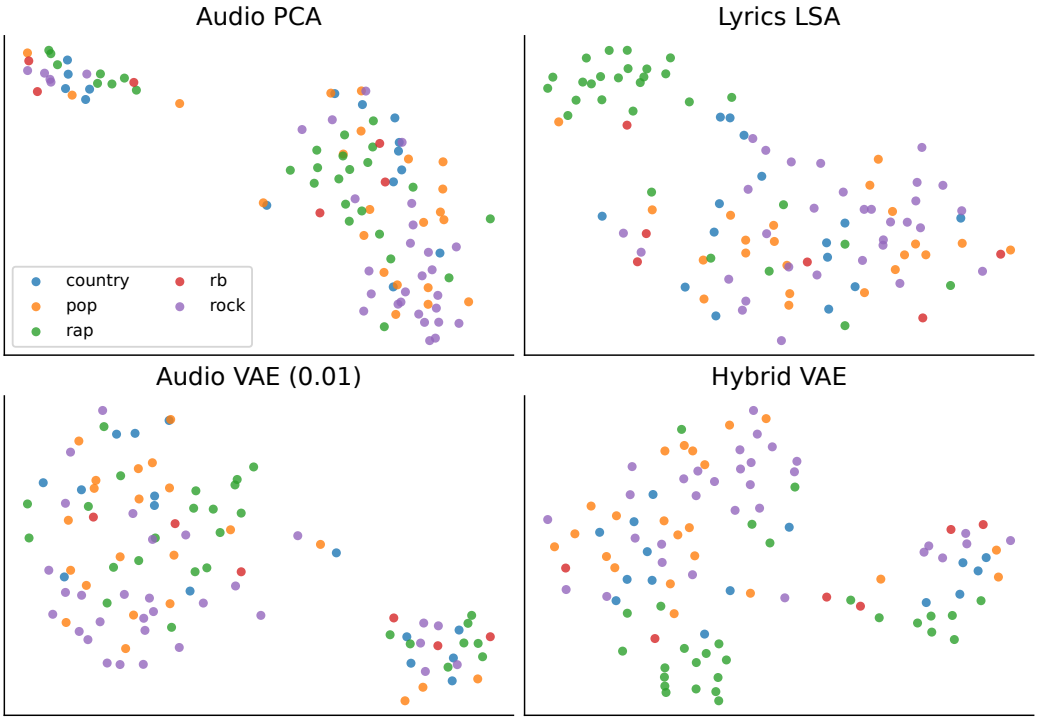


Figure 4: Test-set t-SNE, seed 11, perplexity 20, colored by genre for four representations. Each panel is fitted separately; axes and distances across panels are not comparable.

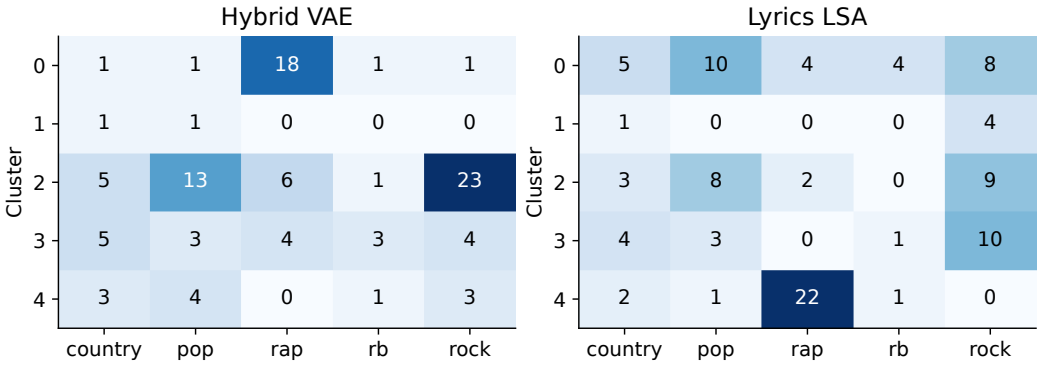


Figure 5: Actual genre counts in the seed-11 held-out clusters. Cluster numbers are arbitrary and cannot be matched across methods by their numeric ID.

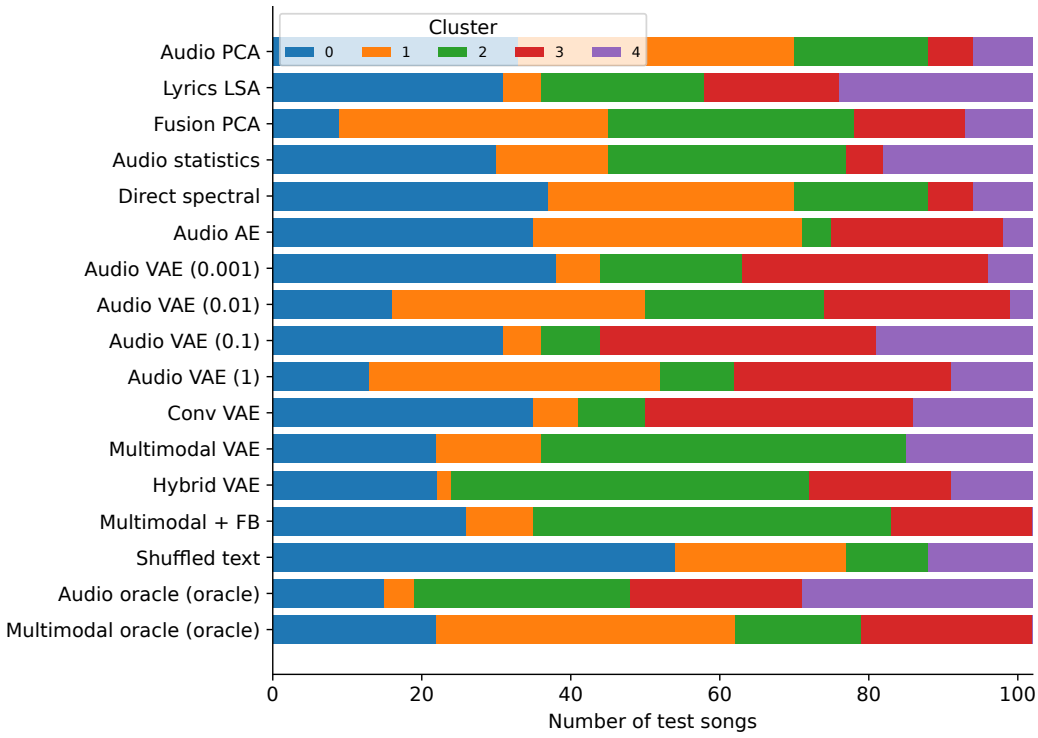


Figure 6: Test cluster sizes for seed 11, including separately identified genre oracles and the shuffled-text control. Small or dominant clusters remain visible.

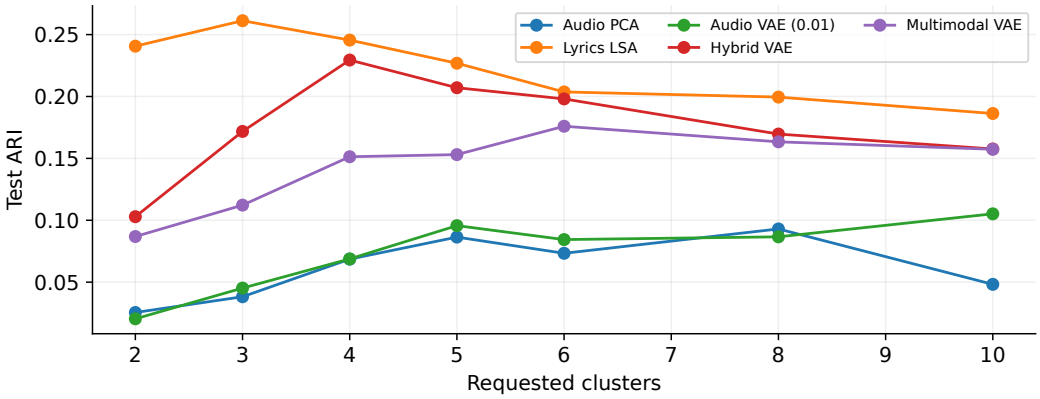


Figure 7: Sensitivity to the prespecified cluster-count grid. Lines show mean test ARI across seeds. The grid is descriptive; no test-based k is selected.

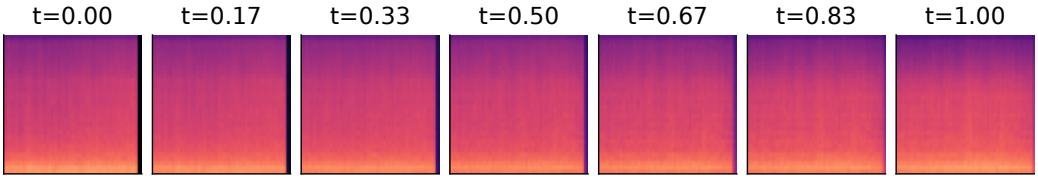


Figure 8: Linear interpolation between two seed-11 audio-VAE posterior means (first test country and rock examples). Smooth decoder output alone does not establish musically meaningful interpolation.

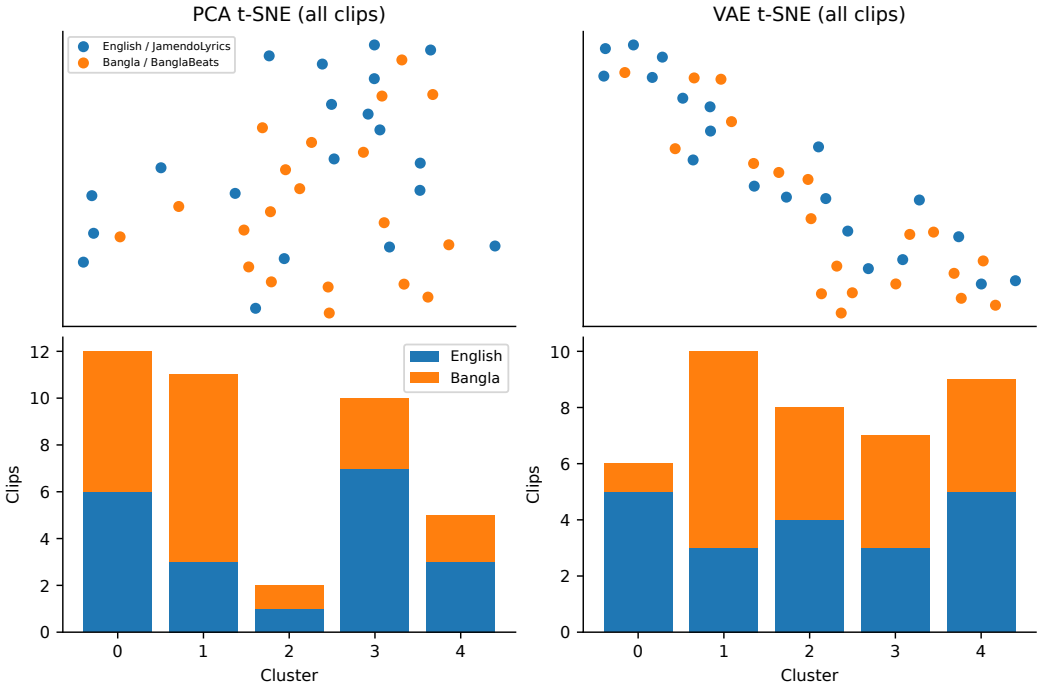


Figure 9: Bilingual course demonstration: t-SNE colored by annotated language/source and actual language counts in the five seed-11 clusters. All 40 clips are included. Bengali original song IDs are unavailable, and language and source cannot be disentangled.

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